DAY 12 OF 30

Load only what changed

Incremental loading.
Faster, cheaper, and easy to get wrong.

Watermark **MERGE** **Change data**

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THE PROBLEM

Reloading everything, every night, stops scaling.

1	**Year one**	The full load takes 10 minutes
2	**Year three**	It takes four hours
3	**The data that changed**	Less than one percent

You are paying to move and rewrite **data that is exactly the same as yesterday**.

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THE SIMPLE VERSION

Read today's newspaper, not the archive.

To catch up on the news, you do not reread every paper ever printed. You read what came out since you last looked.

An incremental load does that: **remember where you stopped, fetch only what is new.**

✓	**Bookmark**	The last point you loaded
✓	**Fetch**	Only rows after it
✓	**Apply**	Add or update, never duplicate

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HOW IT WORKS

The three steps

1. **Find the changes**: a modified date, an ID, or a change feed
2. **Bring only those rows** into the lake
3. **Merge** them: update what exists, insert what is new
4. **Move the bookmark** only after the load succeeds

MERGE is the key

One statement updates matching rows and inserts new ones, as a single transaction.

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FOR ENGINEERS

Where it goes wrong

Deletes are invisible

A watermark on "modified date" never sees a deleted row. You need a change feed or a periodic compare.

Late data

Rows that arrive after the bookmark has moved are skipped unless you reload a safety window.

Reruns must be safe

If a retry inserts the same rows twice, the load is not incremental, it is wrong.

Trust but verify

A scheduled full reconcile catches drift before the business does.

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THE HONEST PART

Incremental or full load?

INCREMENTAL

Large tables with few daily changes

A reliable change marker exists

Loads must finish in a tight window

FULL RELOAD

Small tables that load in seconds

No trustworthy change marker

History gets rewritten often

> **The catch:** an incremental load fails quietly. It still finishes, the row counts look fine, and the table drifts a little further from the source every day.

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FROM REAL WORK

Incremental on a real platform

A data consolidation platform on Databricks for a financial services company.

1. **Over 10 million records** processed each day
2. **Incremental pipelines** and automated loading workflows
3. **Sources** included FTP, AWS S3 and Kafka
4. **dbt tests** enforced data quality on the results

At that volume, full reloads were **never an option**.

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SAVE THIS

Incremental in five lines

1. **Bookmark** where the last load ended
2. **Fetch** only what changed since
3. **MERGE** to update and insert safely
4. **Plan for** deletes and late data
5. **Reconcile** against the source regularly

Tomorrow: Table maintenance. OPTIMIZE, V-Order and VACUUM.

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